Recall

Work from memory, then check with your tutor.

R1. Find $1.4 \div 0.07$.

Answer: _____

R2. Solve $(\frac{2}{5})x = 10$.

Answer: _____

R3. Find the surface area of a closed 2 by 3 by 5 cm box.

Answer: _____

R4. Find $10 - 2.286$.

Answer: _____

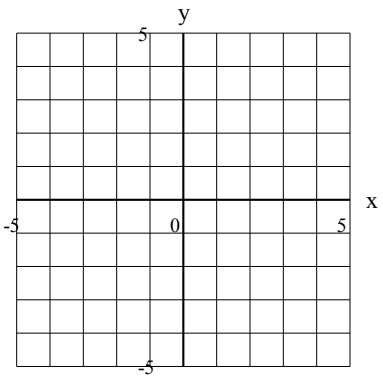
S. Return here near the end of the lesson

A reflection across the y-axis changes __ and keeps __.

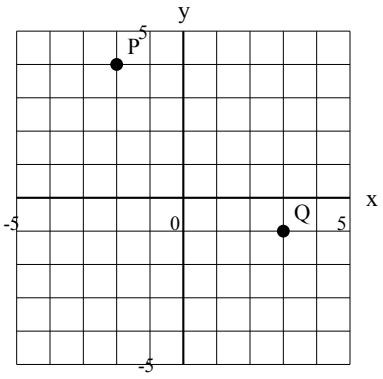
Coordinates and inequalities

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Plot A(2, 3), B(−4, 1), C(−3, −2), and D(1, −4). Name the quadrant of each.



P2. Read P and Q from the grid. State the horizontal move first.



P3. Reflect (−2, 3) across the x-axis. Then reflect the original point across the y-axis.

Answer: _____

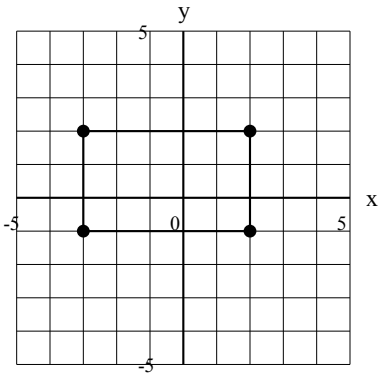
Coordinates and inequalities

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Find the distance from (−2, 3) to (−2, −4), and from (−2, 3) to (5, 3).

Answer: _____

P5. A rectangle has vertices (−3, 2), (2, 2), (2, −1), and (−3, −1). Find its perimeter.



P6. Graph $x > 2$ and $x \geq 2$ on separate number lines. Is 2 a solution of each?

Answer: _____

Coordinates and inequalities

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. A ride requires a height of at least 48 inches. Write an inequality for h. Is 48 included?

Answer: _____

F. Reflect (−5, −2) across the y-axis: (__, __). The distance between the points is __.

Answer: _____

D. Nia writes x > 6 for “at least 6.” She leaves 6 out. Explain and repair her graph.

Answer: _____

Mixed review

M1. Find 1.45 + 3.078.

Answer: _____

M2. Find 3n + 1 when n = 4.

Answer: _____

M3. A rectangle is 8 cm by 2 cm. Find its area.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Reflect A(−4, 1.5) across the y-axis. Give the image and the distance from A to its image.

Answer: _____

Negative numbers, coordinates, and inequalities

The big idea

A signed number gives a position relative to zero. Coordinates use two such positions. Inequalities describe all positions that meet a condition.

Words to know

negative number: A number less than zero. Example: −4 can mean four degrees below zero.

opposites: Numbers the same distance from zero on opposite sides. Example: −6 and 6

absolute value: A number’s distance from zero. Example: $|-6| = 6$

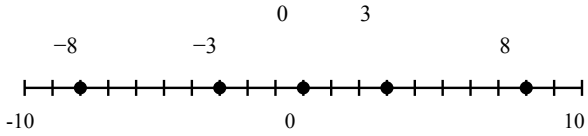
coordinate pair: Two numbers naming horizontal and vertical position. Example: (−2, 3) means left 2, up 3.

reflection: A mirror image across a line. Example: Across the y-axis, (−2, 3) becomes (2, 3).

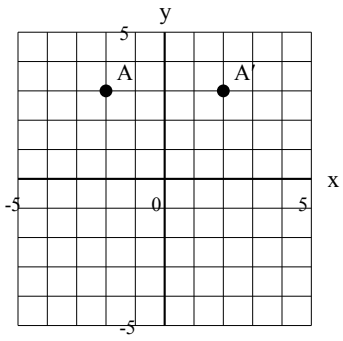
inequality: A comparison using $<$, $>$, $\leq$, or $\geq$. Example: $h \geq 10$ includes 10 and larger numbers.

solution set: All values that make a statement true. Example: $x > 4$ includes 4.5 and 5, but not 4.

How it works



Opposites have equal distances from zero. Numbers increase to the right. Absolute value measures distance from zero, regardless of direction.



The y-axis lies midway between A and A′. A coordinate pair gives horizontal then vertical position. An inequality can describe many solutions.

Worked examples

1. Compare −1.5 and −0.5.

Locate zero, then mark both numbers to its left.

−1.5 lies one unit left of −0.5.

Therefore −1.5 < −0.5, but |−1.5| > |−0.5|.

Answer: −1.5 < −0.5

2. Find the distance from (−1, −3) to (−1, 2).

The x-coordinate stays −1, so the segment is vertical.

From −3 to 0 is 3 units. From 0 to 2 is 2 units.

Total distance = 3 + 2 = 5 units.

Answer: 5 units

Common mistakes

Orders negatives by their digits. Why: Extends positive-number rules unchanged. Fix: Walk rightward on a number line and read the labels.

Gives a negative absolute value. Why: Keeps direction in a distance measure. Fix: Count unit steps from zero.

Says the opposite of −9 is −9. Why: Does not reflect across zero. Fix: Fold a paper number line at zero.

GCF and LCM

The greatest common factor is the largest whole number dividing both numbers. GCF(12, 18) = 6. The least common multiple is the first shared positive multiple. LCM(4, 6) = 12. List factors or multiples to check.

Optional preview

Undo equal steps on both sides. For 2x + 3 = 11, subtract 3 to get 2x = 8. Divide by 2 to get x = 4. A number-line step right adds a positive number.

Try it

1. Order −7, 2, −1 from least to greatest.

2. Order −4, 0, −9 from least to greatest.

3. Reflect (-3, 4) across the y-axis.

4. Reflect (2, -5) across the x-axis.

5. Reflect (-4, -2) across the y-axis.

Answers

1. −7, −1, 2 2. −9, −4, 0 3. (3, 4) 4. (2, 5) 5. (4, −2)

Where this connects

Number lines extend fraction and decimal ideas to negatives. Coordinates will show proportional relationships and data.

Coordinate planes

